

Question number	Answer	Additional guidance		Mark
1(a)	An answer that includes the following points: - glycolysis completed correctly (1) - Krebs cycle completed correctly (1) - Oxidative phosphorylation completed correctly (1)	Accept products written in long hand eg Adenosine Triphosphate.		(3)
		Stage	One other product	
		glycolysis	pyruvate / reduced NAD/NADH	
		Krebs cycle	reduced NAD/NADH/FAD/FADH/ carbon dioxide/CO ₂	
		oxidative phosphorylation	water / H ₂ O/ NAD/FAD	

Question number	Answer	Additional guidance		Mark
1(b)(i)	An answer that includes the following points: - correct number 80 O₂ - correct number 57 CO₂			(2)

Question number	Answer	Additional guidance	
1(c)(i)	Correct answer C A is incorrect as energy is not released from ATP in the conversion of ADP to ATP B is incorrect as energy is not released from ATP in the phosphorylation of ADP D is incorrect as energy is not released from ATP in the loss of adenosine molecules from ATP		(1)

Question number	Answer	Additional guidance	
1(c)(ii)	A calculation showing the following steps: • calculation of energy from 38 ATP (1) • calculation of percentage of energy from glucose (1)	38 x 30.51 (1159.38) (1159.38/2867.48 x 100) = 40.43% Correct answer without working gains 2 marks	(2)

Question number	Answer	Additional guidance	Mark
1(c)(iii)	An answer that includes the following points: • lost as heat (1) • increasing body temperature (1)		(2)

Question number	Answer	Additional guidance	Mark
1(d)	A description that includes two of the following points: - oxygen is terminal electron acceptor / accepts electrons from electron transport chain (1) - forms H₂O / accepts H⁺ from reduced NAD/FAD (1)	Accept electrons cannot be passed along electron transport chain if no O₂ to accept them (1)	(2)